Chapter 1, Problem 23P

Problem

A 1.8-kW electric heater takes 15 min to boil a quantity of water. If this is done once a day and power costs 10 cents/kWh, what is the cost of its operation for 30 days?

Step-by-step solution

Step 1 of 2

Consider the following data:

A 1.8-kW TV receiver tuned for $15\,\text{minutes}$ to boil a quantity of water.

The cost of electricity is, $10\,\text{cents/kWh}$.

Calculate the power consumed for $15\,\text{minutes}$ per day is as follows.

$$\begin{aligned}\text{power consumed for } 15\,\text{minutes per day} &= (1.8\text{-kW})(15\,\text{minutes})\\ &= (1.8\text{-kW})\left(\frac{15}{60}\,\text{hours}\right)\\ &= \frac{27}{60}\,\text{Wh}\\ &= 0.45\,\text{kWh}\end{aligned}$$

Calculate the power consumed for 30 days is as follows.

$$\begin{aligned}\text{power consumed for } 30 \text{ days} &= (0.45\,\text{kWh})(30)\\ &= 13.5\,\text{kWh}\end{aligned}$$

Step 2 of 2

Calculate the cost of electricity is as follows.

$$\begin{aligned}13.5\,\text{kWh @ } 10\,\text{cents/kWh} &= (13.5\,\text{kWh})(10\,\text{cents/kWh})\\ &= 135\,\text{cents}\\ &= \frac{135}{100}\,\text{dollars} \quad (\text{since } 1\,\text{dollar} = 100 \text{ cents})\\ &= \$1.35\end{aligned}$$

Therefore, the cost of operation for 30 days is $\boxed{\$1.35}$.